

Education

- 2020–2022 **Seoul Science High School**, Seoul, Republic of Korea
2023–Present **B.S. in Mathematical Sciences**, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea, Double Major in Computer Science

Course Taken

- Mathematical Science Introduction to Number Theory, Linear Algebra, Discrete Mathematics, Analysis I, II, Modern Algebra I, II, Topology, Combinatorial Topology, Complex Variables I, Elementary Probability Theory, Lebesgue Integral Theory
- Computer Science Data Structure, Introduction to Algorithms, Programming Principles, System Programming
- Theoretical Computer Science Algorithmic Graph Theory, Formal Languages and Automata, Algorithms Design and Analysis for NP-Hard Problems, Seminar in Theoretical Computer Science, Design and Analysis of Algorithm

Research Interests

Structural Graph Theory, Graph Algorithms, Parameterized Complexity, Approximation Algorithms, Computational Complexity

Research Experience

- Jun. 2025–
Dec. 2025 **Individual Study**, Advisor: Prof. Sebastian Wiederrecht, Topic: Graphs of mim-width 1
- Studied graph classes of mim-width 1.
 - Investigated cographs, permutation graphs, and distance-hereditary graphs.
 - Explored perfect contractibility.
- Jan. 2026–
Jun. 2026 **Undergraduate Research Program**, Advisor: Prof. Sebastian Wiederrecht, Topic: Rooted Minors in Planar Graphs and Decompositions with Small Face Covers
- Studied rooted minors in planar graphs.
 - Investigated sphere-cut decompositions.
 - Surveyed structural graph theory.
 - Wrote a final technical report.
- Jul. 2026–
Present **Individual Study**, Advisor: Dr. Roohani Sharma, Topic: Multiway cut on Directed graph with many parameters

Teaching Experience

- Summer 2024 **Teaching Assistant**, Korean Mathematical Olympiad(KMO) Summer School
- Fall 2025 **Teaching Assistant**, Algorithmic Graph Structure Theory, KAIST
- Spring 2026 **Teaching Assistant**, Algorithmic Graph Theory, KAIST
- Spring 2026 **Participant**, Formal Language and Automata Autograder Project, KAIST

Honors and Awards

- 2022 International Mathematical Olympiad (IMO) National Finalist, Republic of Korea

2024 Spring Department Valedictorian, Department Honors Scholarship, KAIST Mathematical Science
2024 Spring Dean's List, KAIST Mathematical Science
2024, 2025 Silver Prize, Undergraduate Mathematics Competition, Korean Mathematical Society
2025–present KAIST Presidential Fellowship(KPF), from the President of KAIST
2025–present National Science and Technology Scholarship

■ Conferences Attended

Jul. 2025 2025 Summer School on Combinatorics and Algorithms (*Summer School Webpage*)
Jan. 2026 Workshop on Algorithm for Students 2026 (*Workshop Webpage*)
Aug. 2026 2026 Summer School on Combinatorics and Algorithms (*Summer School Webpage*)

■ Leadership

2023 **Tournament Director**, Pannorm, KAIST Board Game Club
2024 **President**, *KAIST Mathematical Problem Solving Club*
2025 **President**, **The Science and Technology Division**, *KAIST Student Clubs Association*

■ Skills

Programming Python, C/C++, LaTeX
Languages Korean (Native), English